

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	3	3	3	4	2	4	2	3	1	2	2	3	3	2	1	4	2	4	1	3	4	4	2	2	3	4	3	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	1	2	1	3	1	2	3	1	1	3	4	3	3	1	2	2	1	3	1	3	4	1	3	1	2	3	3	2	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	3	4	1	2	3	4	1	1	2	4	1	1	4	2	1	1	2	4	2	4	4	3	1	1	4	1	1	3	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	2	3	3	1	1	2	3	2	1	4	2	4	3	4	3	1	1	2	2	3	3	3	1	2	3	3	1	1	3	1
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	1	4	2	4	4	4	4	2	2	2	2	3	3	3	3	3	1	2	1	1	3	1	1	2	4	3	3	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	2	3	3	1	2	3	1	4	2	1	4	3	1	3	3	4	2	4	2	1	4	4	4	2	2	3	1	4	2	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	3	4	4	3	1	4	1	2	3	1	2	4	3	4	3	4	2	1	2	3										

HINT - SHEET

SUBJECT : PHYSICS

SECTION - A

1. **Ans (4)**

$$V_D - V_C - iR = 0$$

$$0 - V_C - 1(2) = 0$$

$$V_C = -2V$$

2. **Ans (4)**

$$i_1 (R_1 + r) = i_2 (R_2 + r)$$

$$1(2 + r) = 0.25 (11 + r)$$

$$r = 1\Omega$$

3. **Ans (3)**

$$V = ir = \frac{\varepsilon}{R + r}r$$

4. **Ans (3)**

$$C_O = \frac{\varepsilon_0 A}{d}$$

$$C_K = \frac{\varepsilon_0 A}{d - t + \frac{t}{k}} = \frac{\varepsilon_0 A}{d - \frac{d}{3} + \frac{d}{3 \times 2}}$$

$$= \frac{6}{5} \frac{\varepsilon_0 A}{d} = \frac{6}{5} C_O$$

5. **Ans (3)**

$$\text{At } r = R$$

$$E = \frac{kQ}{R^2}$$

$$\text{At } r = 3R/2$$

$$E' = \frac{kQ}{(3R/2)^2} = \frac{4}{9} \frac{kQ}{R^2} = \frac{4E}{9}$$

6. **Ans (4)**

$$\text{Time period } T = 2\pi \sqrt{\frac{M}{K}}$$

$$T' = 2\pi \sqrt{\frac{M + \frac{M}{2}}{K}} = 2\pi \sqrt{\frac{3M}{2K}}$$

$$\therefore T' = \sqrt{\frac{3}{2}} T$$

7. **Ans (2)**

$$\frac{Q}{t} = \frac{kA(T_1 - T_2)}{L} \propto (T_1 - T_2)$$

$$\frac{4}{(Q/t)_2} = \frac{220 - 200}{440 - 400}$$

$$\left(\frac{Q}{t}\right)_2 = 8 \text{ J/s}$$

8. **Ans (4)**

Here $\vec{F} = 4\hat{i} + 5\hat{j} - 6\hat{k}$ and

$$\vec{r} = (2-2)\hat{i} + (0-(-2))\hat{j} + (-3-(-2))\hat{k} = 2\hat{j} - 1\hat{k}$$

$$\vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & -1 \\ 4 & 5 & -6 \end{vmatrix} = -7\hat{i} - 4\hat{j} - 8\hat{k}$$

9. **Ans (2)**

$$I = \frac{5}{6} \cdot \frac{MR^2}{2} = \frac{5}{12} MR^2$$

10. **Ans (3)**

$$Y = \frac{F\ell}{A\Delta\ell} \Rightarrow \Delta\ell = \frac{F\ell}{AY}$$

$$(\Delta\ell)_{\text{Steel}} = (\Delta\ell)_{\text{Brass}}$$

$$\Rightarrow \frac{W_S\ell}{AY_S} = \frac{W_B\ell}{AY_B}$$

$$\Rightarrow \frac{W_S}{W_B} = \frac{Y_S}{Y_B} = \frac{2}{1}$$

11. **Ans (1)**

$$T = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{GM}} \sqrt{r} \quad \left(\text{as } v = \sqrt{\frac{GM}{r}}\right)$$

$$T = \frac{2\pi}{\sqrt{GM}} r^{3/2} \Rightarrow T^2 = \frac{4\pi^2}{GM} \cdot r^3$$

$$\text{Comparing } K = \frac{4\pi^2}{GM}$$

12. **Ans (2)**

$$\text{Escape Velocity} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2G}{R} \cdot \left(\frac{4}{3}\pi R^3 \rho\right)}$$

$$\propto R\sqrt{\rho}$$

$$\therefore \text{Ratio, } \frac{v_e}{v_p} = 1 : 2\sqrt{2}$$

13. **Ans (2)**

$$x_{\text{cm}} = \frac{2M \times 0 + M(2\sqrt{2}) + M \times 0}{4M} = \frac{1}{\sqrt{2}}$$

$$y_{\text{cm}} = \frac{2M \times 0 + M \times 0 + M(2\sqrt{2})}{4M} = \frac{1}{\sqrt{2}}$$

$$\vec{r} = \frac{1}{\sqrt{2}} (\hat{i} + \hat{j})$$

14. **Ans (3)**

$$V_0^2 = 0^2 + 2(\alpha r)(n2\pi r)$$

$$\Rightarrow \alpha = \frac{V_0^2}{4\pi nr^2}$$

15. **Ans (3)**

$$W = \int_0^2 F dy = \int_0^2 (10y + 3y^2) dy$$

$$= [5y^2 + y^3]_0^2 = 20 + 8 = 28 \text{ J}$$

16. **Ans (2)**

Length of the air column,

$$\ell = 40 \text{ cm} = 40 \times 10^{-2} \text{ m}$$

Frequency of tuning fork, $n = 450 \text{ Hz}$

In resonance, the frequency of tuning fork will be equal to frequency of air column.

$$n = \frac{v}{4\ell} \quad (\text{where, } v \text{ is the velocity of sound})$$

$$\text{or } V = n \times 4\ell = 450 \times 4 \times 0.4 \text{ ms}^{-1}$$

$$= 720 \text{ ms}^{-1}$$

17. Ans (1)

The standard equation to wave is

$$y = a \sin(kx - \omega t) \quad \dots\dots(i)$$

Where, ω is angular velocity and t is time.

Given, equation is

$$y = a \sin(0.01x - 2t) \quad \dots\dots(ii)$$

Comparing the given equation with standard waveform, we get

$$kx = 0.01x \text{ and } \omega t = 2t$$

$$\therefore k = 0.01 \text{ and } \omega = 2$$

$$\text{Thus, velocity} = \frac{\omega}{k} = \frac{2}{0.01} = 200 \text{ cm/s}$$

$\therefore$ Velocity of propagation of the wave is 200 cm/s.

18. Ans (4)

$$X = \frac{A^{1/2} B^2}{C^3 D^{1/3}}$$

$$\frac{\Delta x}{x} = \frac{1}{2} \frac{\Delta A}{A} + \frac{2\Delta B}{B} + \frac{3\Delta C}{C} + \frac{1}{3} \frac{\Delta D}{D}$$

$$\% \text{ change} = \frac{1}{2} \times 2\% + 2 \times 1 + 3 \times 3 + \frac{1}{3} \times 3$$

$$= 1 + 2 + 9 + 1$$

$$= 13\%$$

19. Ans (2)

$$a = \frac{1 \times 10}{1+1} = 5 \text{ m/s}^2$$

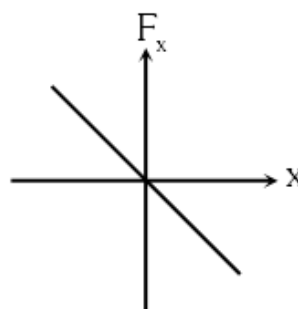
$$s = ut + \frac{1}{2} at^2 = 0 + \frac{1}{2} \times 5 \times 2^2 = 10 \text{ m}$$

20. Ans (4)

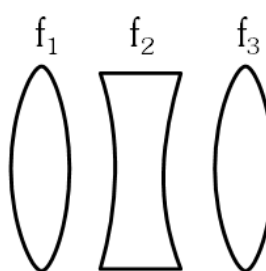
Restoring force of spring is

$$F = -kx$$

$$F \propto -x$$



22. Ans (3)



$$f_1 = f_3$$

$$f_2 = -25 \text{ cm}$$

$$f_1 = f_3 = 40 \text{ cm}$$

$$\therefore \frac{1}{f_{\text{net}}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \text{ and } P = \frac{1}{f_{\text{net}}(\text{m})}$$

$$\therefore \frac{1}{f_{\text{net}}} = \frac{1}{40} - \frac{1}{25} + \frac{1}{40}$$

$$= \frac{2}{40} - \frac{1}{25} = \frac{1}{100 \text{ cm}}$$

$$\therefore P = \frac{100}{100} = 1 \text{ D or } f = \frac{1}{P} = 1 \text{ m} = 100 \text{ cm}$$

23. Ans (4)

According to mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad v = \frac{uf}{u-f}$$

$$\text{Again, } \frac{1}{v'} + \frac{1}{(u+L)} = \frac{1}{f}$$

$$v' = \frac{(u+L)f}{u+L-f}$$

$$L' = v - v' = \frac{uf}{u-f} - \frac{(u+L)f}{u+L-f} = \frac{f^2 L}{(u-f)(u+L-f)}$$

Since the object is short, i.e., $u \gg L$

$$\therefore L' = L \left(\frac{f}{u-f} \right)^2$$

24. Ans (4)

Fringes are formed at same spot means fringes coincide

$$n_1 \lambda_1 = n_2 \lambda_2$$

$$(n + 4) 5200 \text{Å} = (n) 7800 \text{Å}$$

$$(n + 4)4 = 6n$$

$$n = 8$$

25. Ans (2)

$$I = I_m \cos^2 \frac{\phi}{2}$$

$$\frac{I_m}{2} = I_m \cos^2 \frac{\phi}{2} \Rightarrow \phi = 90^\circ$$

$$\Delta x = \frac{\lambda}{4}$$

$$y = \frac{D}{d} \times \Delta x = \frac{D}{d} \times \frac{\lambda}{4}$$

26. Ans (2)

For electron

$$\lambda = \frac{12.27}{\sqrt{V}} \text{Å}$$

$$V_1 = 25 \text{ kV} \quad \lambda_1 = 1$$

$$V_2 = 100 \text{ kV} \quad \lambda_2 = 1'$$

$$\frac{\lambda'}{\lambda} = \sqrt{\frac{25 \text{ kV}}{100 \text{ kV}}} = \frac{1}{2} \Rightarrow \lambda' = \frac{1}{2} \lambda$$

27. Ans (3)

$$P = 200 \text{ W}, \lambda = 0.6 \text{ mm}$$

$$n = 5 \times 10^{24} \times 200 \times 0.6 \times 10^{-6}$$

$$n = 6 \times 10^{20}$$

but efficiency is 25% so

$$n' = \frac{25}{100} \times 6 \times 10^{20} = 1.5 \times 10^{20}$$

28. Ans (4)

$$eV = \frac{hc}{\lambda} - \frac{hc}{\lambda_0} \quad \dots(i)$$

$$\frac{eV}{4} = \frac{hc}{2\lambda} - \frac{hc}{\lambda_0} \quad \dots(ii)$$

From equation (i) and (ii)

$$\Rightarrow 4 = \frac{\frac{1}{\lambda} - \frac{1}{\lambda_0}}{\frac{1}{2\lambda} - \frac{1}{\lambda_0}} \quad \text{On solving } \lambda_0 = 3\lambda$$

29. Ans (3)

In (a) & (c) circuits, both the junctions are in same biasing conditions so offers equal resistances.

Since both are in series, therefore equal potential will drop across the junction.

30. Ans (4)

$$y = \overline{(A + B)} + \overline{(A + B)} = \overline{(A + B)} \cdot \overline{(A + B)}$$

(By De'Morgans theorem)

$$= (A + B) \cdot (A + B)$$

$$= A + B \Rightarrow \text{OR Gate}$$

31. Ans (3)

For zener diode $\rightarrow$ Doping is high & Depletion region is thin & It is operated in Reverse Bias region & Zener voltage (V_z) is constant

33. Ans (2)

$$I = \frac{V_0}{R} \left(1 - e^{-\frac{R}{L}t} \right) = \frac{1}{1.5} \left(1 - e^{-\frac{1.5}{3} \times 2} \right)$$

$$= \frac{1}{3} (1 - 0.37) = 0.21 \text{ A}$$

34. Ans (1)

$$E_{\text{peak}} = c B_{\text{peak}}$$

$$= 3 \times 10^8 \times 2 \times 4\pi \times 10^{-7}$$

$$= 753 \text{ volt/m}$$

$$\vec{S} = \frac{1}{\mu_0} (\vec{E} \times \vec{B})$$

Since $\vec{B}$ is in $-\hat{i}$ & wave is travelling towards $\hat{k}$,
 $\therefore \vec{E}$ is towards $\hat{j}$
 $\therefore$ Equation of electric field
 $\vec{E} = 753 \cos(\omega t - kz) \hat{j}$

35. Ans (3)

Susceptibility $\chi = 99$

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi$$

$$\mu = \mu_0 (1 + \chi)$$

$$= 4\pi \times 10^{-7} [1 + 99]$$

$$= 4\pi \times 10^{-5}$$

SECTION - B

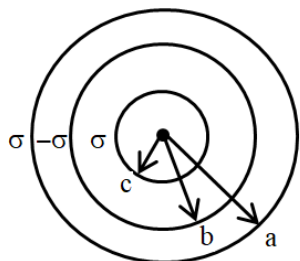
36. Ans (1)

$$P = \frac{V^2}{R_{\text{eq}}} \Rightarrow R_{\text{eq}} = \frac{(10)^2}{25} = 4\Omega$$

$$\Rightarrow \left(\frac{1}{R} + \frac{1}{6} \right)^{-1} = 4\Omega$$

$$\Rightarrow R = 12\Omega$$

37. Ans (2)



$$V_A = k4\pi\sigma \left[\frac{a^2 - b^2 + c^2}{a} \right]$$

$$= k4\pi\sigma (2C)$$

$$V_B = k4\pi\sigma \left[\frac{a^2}{a} + \frac{-b^2 + c^2}{b} \right]$$

$$= k4\pi\sigma \left[\frac{ac}{b} \right]$$

$$V_C = k4\pi\sigma \left[\frac{a^2}{a} - \frac{b^2}{b} + \frac{c^2}{c} \right] = k4\pi\sigma (2C)$$

38. Ans (3)

$$\eta = \frac{1}{1 + \beta} \Rightarrow \frac{20}{100} = \frac{1}{1 + \beta} \Rightarrow \beta = 4$$

$$\beta = \frac{Q_2}{W} \Rightarrow 4 = \frac{Q_2}{20}$$

$$\Rightarrow Q_2 = 80 \text{ J}$$

39. Ans (1)

$$\text{COP} = \frac{T_L}{T_H - T_L}$$

$$15 = \frac{240}{T_H - 240}$$

$$T_H = 256 \text{ K} = -17^\circ\text{C}$$

40. Ans (1)

$$\frac{P}{\rho} = \frac{RT}{M_w} \Rightarrow \rho = \frac{PM_w}{RT}$$

$$\rho = \frac{498 \times 10^3 \times 2 \times 10^{-3}}{8.3 \times 400}$$

$$\rho = 0.3 \text{ kg/m}^3$$

41. Ans (3)

According to wein's law, $\lambda_m \propto \frac{1}{T}$ when temperature is increased, the peak emission wavelength of a black body shifts to lower wavelength.

42. Ans (4)

Centre of mass may lie on centre of gravity net torque of gravitational pull is zero about centre of mass.

$$\text{Mechanical advantage} = \frac{\text{Load}}{\text{Effort}} > 1$$

$$\Rightarrow \text{Load} > \text{Effort}$$

43. Ans (3)

$$mgh = \frac{1}{2} mv^2 (1 + K^2/R^2)$$

$$\Rightarrow 10 h = \frac{1}{2} (100) \left(1 + \frac{1}{2}\right)$$

$$\Rightarrow h = \frac{10 \times 3}{4} = \frac{15}{2} \text{ m}$$

$$S = \frac{h}{\sin 30^\circ} = \frac{15}{2} \cdot \frac{2}{1} = 15 \text{ m}$$

44. Ans (3)

As surface area decreases so energy is released.

Released energy

$$= 4\pi R^2 T [n^{1/3} - 1] \quad \text{where } R = n^{1/3} r$$

$$= 4\pi R^3 T \left[\frac{1}{r} - \frac{1}{R} \right] = 3VT \left[\frac{1}{r} - \frac{1}{R} \right]$$

45. Ans (1)

$$\text{Gravitational constant} = [M^{-1}L^3T^{-2}]$$

$$\text{Escape velocity} = [LT^{-1}]$$

$$\text{Gravitational potential} = [L^2T^{-2}]$$

$$\text{Acceleration due to gravity} = [LT^{-2}]$$

46. Ans (2)

$$\text{Power} = 10 \times 10^3 \text{ W} = 10^4 \text{ J/s}$$

Amount of U^{235} to operate 10 kW reactor is

$$= \frac{10^4 \times 235}{6.02 \times 10^{23} \times 200 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$= 1.22 \times 10^{-7} \text{ g/s}$$

47. Ans (2)

$$f_0 = 20 \text{ m} = 2000 \text{ cm}$$

$$f_e = 2 \text{ cm}$$

for Normal adjustment

$$\rightarrow \text{M.P.} = -\frac{f_0}{f_e} = \frac{-2000}{2} = -1000$$

$$\rightarrow \text{Distance between both lens} = f_0 + f_e$$

$$= 2000 + 2$$

$$= 2002 \text{ cm}$$

$$= 20.02 \text{ m}$$

$\rightarrow$ Image is inverted and magnified

$\rightarrow$ Aperture of eye piece is smaller than objective.

48. Ans (1)

Statement (C) is correct because, the magnetic field outside the toroid is zero and they form closed loops inside the toroid itself.

Statement (E) is correct because we know that super conductors are materials inside which the net magnetic field is always zero and they are perfect diamagnetic.

$$\mu_r = 1 + \chi$$

$$\chi = -1$$

$$\mu_r = 0$$

49. Ans (3)

$$\text{Pitch} = 0.5 \text{ mm}, \text{LC} = \frac{0.5 \text{ mm}}{100} = 0.005 \text{ mm}$$

$$(1) \text{ ZE} : \Rightarrow +ve$$

$$\text{ZE} = +2 \text{ div} \times 0.005 \text{ mm} = 0.010 \text{ mm}$$

(2) Measurement :

$$\text{MSR} = 0.5 \text{ mm} \times 8 = 4.0 \text{ mm}$$

$$\text{CSR} = 83 \times 0.005 \text{ mm} = 0.415 \text{ mm}$$

$$\text{Corrected reading} = 4.0 \text{ mm} + 0.415 \text{ mm} -$$

$$(+0.010) \text{ mm}$$

$$= 4.405 \text{ mm.}$$

50. Ans (1)

$$\text{Figure of merit } K = \frac{E}{(R + G)} \cdot \frac{1}{\theta}$$

$$K_1 = \frac{2}{(5080)} \cdot \frac{1}{20} = 1.9 \times 10^{-5}$$

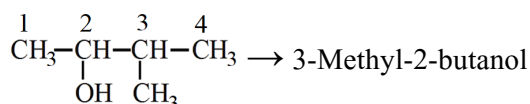
$$K_2 = \frac{4}{(10080)} \cdot \frac{1}{20} = 1.9 \times 10^{-5}$$

$$K = \frac{K_1 + K_2}{2} = 1.9 \times 10^{-5}$$

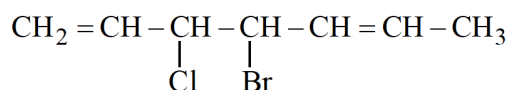
SUBJECT : CHEMISTRY

SECTION-A

51. Ans (3)



52. Ans (4)



No. of SI = 2^n | $n = 3$ (Stereo centre)

$$= 2^3 = 8$$

54. Ans (3)

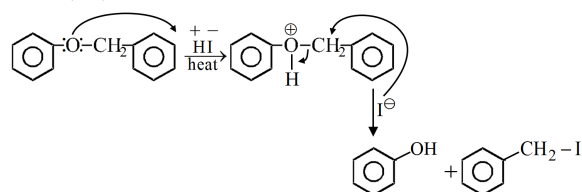
Due to Resonance + Hyperconjugation

55. Ans (1)

Stability order :- Aromatic > Non Aromatic >

Anti Aromatic

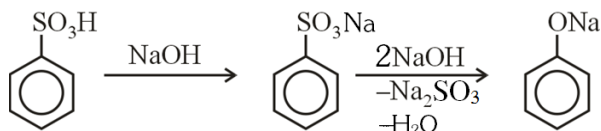
56. Ans (2)



57. Ans (3)

Tertiary alcohol immediately gives cloudiness with leucas reagent.

58. Ans (3)



61. Ans (3)

NCERT XII, Table 12.2, Pg # 360

69. Ans (2)

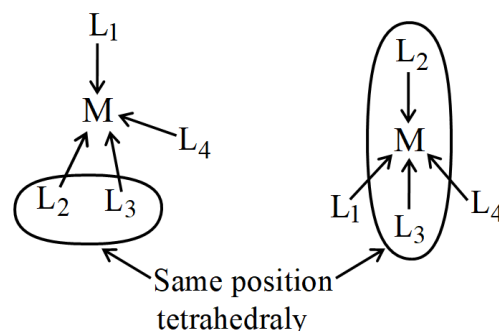
PH₃ has least MP & BP due to no H bond wrt NH₃ &

Less molar weight wrt least 15th group hydrides.

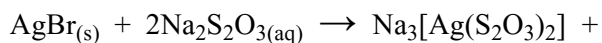
70. Ans (4)

Fact

71. Ans (1)

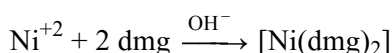


72. Ans (1)



NaBr

73. Ans (4)



74. Ans (2)

$$\begin{aligned} \frac{d[\text{SO}_3]}{dt} &= \frac{40}{80} \text{ mol min}^{-1} = \frac{1}{2} \text{ mol min}^{-1} \\ \frac{-d[\text{O}_2]}{dt} &= \frac{1}{2} \times \frac{d[\text{SO}_3]}{dt} = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} \text{ mol min}^{-1} \\ &= 8 \text{ g min}^{-1} \end{aligned}$$

75. Ans (1)

$$r \propto [C]^1 \quad [\text{For first order reaction}]$$

$$\frac{r_1}{r_2} = \frac{C_1}{C_2} = \frac{3 \times 10^{-4}}{2 \times 10^{-4}} = \frac{3}{2}$$

$$k = \frac{2.303}{t_{75} - t_{25}} \log \frac{C_1}{C_2}$$

$$k = 0.008290 \text{ min}^{-1}$$

$$t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.008290}$$

$$= 83.33 \text{ min}$$

76. Ans (1)

$$P_A^\circ \left(\frac{1}{1+3} \right) + P_B^\circ \left(\frac{3}{1+3} \right) = 550$$

$$\Rightarrow 0.25 P_A^\circ + 0.75 P_B^\circ = 550 \text{ mm Hg} \quad \dots(1)$$

On adding 1 mole B

$$P_A^\circ \left(\frac{1}{1+4} \right) + P_B^\circ \left(\frac{4}{1+4} \right) = 560$$

$$\Rightarrow 0.20 P_A^\circ + 0.8 P_B^\circ = 560 \text{ mm Hg} \quad \dots(2)$$

On solving eqⁿ (1) and eqⁿ (2)

$$P_A^\circ = 400 \text{ mm Hg} \quad P_B^\circ = 600 \text{ mm Hg}$$

77. Ans (2)

NCERT XII, Part-I Pg # 74 (Edition, 2017)

78. Ans (4)

$$\text{Reducing power} \propto \frac{1}{\text{SRP}}$$

79. Ans (2)

NCERT (2021), Part-I, Pg # 32

80. Ans (4)

NCERT (2021), Part-I, Pg # 198

81. Ans (4)

NCERT (2021), Part-I, Pg # 189

82. Ans (3)

NCERT (2021), Part-I, Pg # 175

83. Ans (1)

NCERT (2021), Part-I, Pg # 17

84. Ans (1)

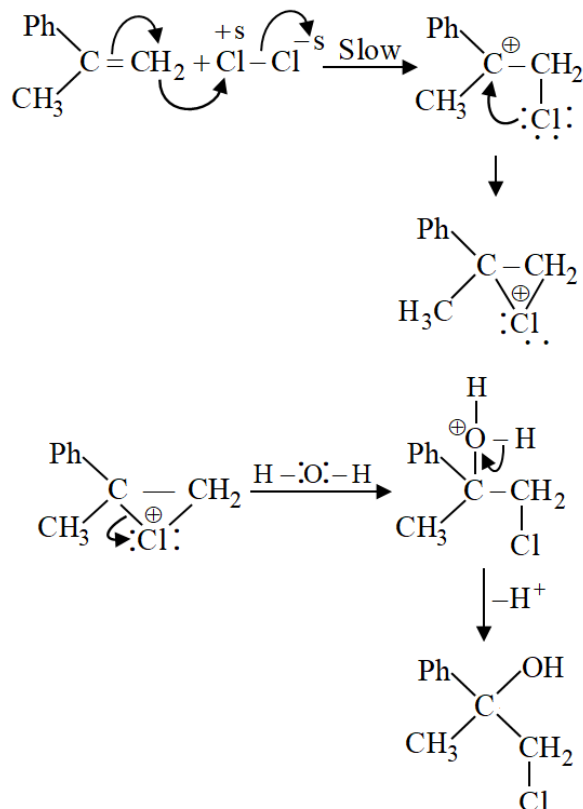
NCERT (2021), Part-I, Pg # 209

85. Ans (4)

NCERT (2021), Part-I, Pg # 169

SECTION-B

87. Ans (1)



90. Ans (1)

$$\begin{aligned} \%Cl &= \frac{35.5}{143.5} \times \frac{W_{\text{AgCl}}}{W_{\text{org.comp}}} \times 100 \\ &= \frac{35.5}{143.5} \times \frac{0.5740}{0.3780} \times 100 = 37.57\% \end{aligned}$$

94. Ans (1)

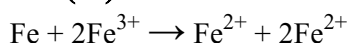
Fact for vanadium, Zn has fulfilled confi.

96. Ans (2)

$$\pi = icRT$$

$$i = \frac{0.369}{0.01 \times 0.082 \times 300} = 1.5$$

98. Ans (2)



$$n = 2$$

$$\Delta G^\circ = -nFE_{\text{cell}}^\circ$$

$$= -2 \times 96500 \times 1.2$$

$$= -231600 \text{ J}$$

$$= -231.6 \text{ kJ}$$

99. **Ans (1)**
NCERT (2021), Part-I, Pg # 221

100. **Ans (4)**
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101. **Ans (2)**
XI NEW NCERT Pg.# 8

102. **Ans (4)**
NCERT (XI) Pg. # 20

103. **Ans (3)**
NCERT XI, Pg.# 36

104. **Ans (4)**
NCERT (XI) Pg. # 42

105. **Ans (3)**
NCERT XI Pg. # 78

106. **Ans (1)**
NCERT-XII Pg#31

107. **Ans (1)**
NCERT XI Pg. # 215

108. **Ans (2)**
NCERT-11th Pg # 229

109. **Ans (2)**
NCERT-11th Pg # 230

110. **Ans (3)**
NCERT-XI, Pg. # 249

111. **Ans (3)**
NCERT-XII Pg. # 225

112. **Ans (3)**
NCERT XII Pg. # 262

113. **Ans (1)**
NCERT-XII Pg#153

114. **Ans (2)**
NCERT-XII Pg#83

115. **Ans (3)**
NCERT-XII Pg#88

116. **Ans (3)**
NCERT-XII Pg#90

117. **Ans (1)**
NCERT XII Pg. No. # 264

118. **Ans (1)**
NCERT XII Pg. No. # 266

119. **Ans (3)**
NCERT XII Pg. No. # 244

120. **Ans (1)**
NCERT XII Pg. No. # 244

121. **Ans (4)**
NCERT XII Pg. No. # 233

122. **Ans (1)**
NCERT XII Pg. No. # 232

123. **Ans (1)**
NCERT XII Pg. No. # 95

124. **Ans (4)**
NCERT-XII Pg#106

125. **Ans (2)**
NCERT-XII Pg#90

126. **Ans (4)**
NCERT-XII Pg#76

127. **Ans (4)**
NCERT XI Pg # 38

128. **Ans (4)**
NCERT XI Pg # 24

129. **Ans (4)**
NCERT XI Pg. No. # 32,33

130. **Ans (2)**
NCERT XI Pg. No. # 85, 86

131. **Ans (2)**
NCERT XI Pg. No. # 93, 94, 95, 97

132. **Ans (2)**
NCERT XI Pg. No. # 75

133. **Ans (2)**
NCERT XI Pg. No. # 219

134. **Ans (3)**
NCERT XI Pg. No. # 248, 249

135. **Ans (3)**
NCERT XI Pg. No. # 233

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136. **Ans (3)**
XI NEW NCERT Pg.# 5

137. **Ans (3)**
NCERT XI, Page # 27

138. **Ans (3)**
XI NCERT Page # 40

139. **Ans (1)**
NCERT-11th Pg # 212

140. **Ans (2)**
NCERT-12th

141. **Ans (1)**
NCERT XII Pg. # 235

142. **Ans (1)**
NCERT-XII Pg#81

143. **Ans (3)**
NCERT-XII Pg#95

144. **Ans (1)**
NCERT XII Pg. No. # 267

145. **Ans (1)**
NCERT-XII Pg#193

146. **Ans (2)**
NCERT-XII Pg#61

147. **Ans (4)**
NCERT XI Pg # 96,97

148. **Ans (3)**
Lady finger, Cotton, Chinrose

149. **Ans (3)**
NCERT XI Pg. No. # 213

150. **Ans (4)**
NCERT XI Pg. No. # 246

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SECTION - A

151. **Ans (2)**
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154. **Ans (1)**
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155. **Ans (2)**
NCERT Pg # 312

156. **Ans (3)**
NCERT Pg # 51-52

157. **Ans (1)**
NCERT Pg # 63

158. **Ans (4)**
NCERT Pg # 310, 311

159. **Ans (2)**
NCERT Pg # 53

160. **Ans (1)**
NCERT Pg # 90, 91

161. **Ans (4)**
NCERT Pg # 149

167. **Ans (2)**
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168. **Ans (4)**
NCERT Page No. # 339

169. **Ans (2)**
NCERT Page No. # 50

170. **Ans (1)**
NCERT Page No. # 272-273

171. **Ans (4)**
NCERT Page No. # 111-115

172. **Ans (4)**
NCERT Page No. # 58-59

173. **Ans (4)**
NCERT Page No. # 46-48, 50-51
174. **Ans (2)**
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175. **Ans (2)**
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176. **Ans (3)**
NCERT XI, Pg. # 121
177. **Ans (1)**
NCERT XI Pg. # 121 Fig 10.1
178. **Ans (4)**
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179. **Ans (2)**
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180. **Ans (3)**
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181. **Ans (3)**
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182. **Ans (4)**
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183. **Ans (4)**
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184. **Ans (3)**
NCERT-XI Pg#106

185. **Ans (1)**
NCERT-XI Pg#106

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187. **Ans (1)**
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188. **Ans (2)**
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189. **Ans (3)**
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190. **Ans (1)**
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191. **Ans (2)**
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196. **Ans (4)**
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197. **Ans (2)**
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198. **Ans (1)**
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199. **Ans (2)**
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200. **Ans (3)**
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